

SQL Project Report: E-Commerce Order Analysis

1. Introduction

This project analyzes e-commerce order transactions using SQL Server Management Studio (SSMS). SQL querying, filtering, aggregation, and sorting techniques were applied to evaluate sales performance, customer purchasing behavior, payment trends, referral channel effectiveness, and operational performance. The analysis also included outlier identification and high-value order assessment to uncover actionable business insights and support data-driven decision-making.

2. Business Problem

The e-commerce business generates a large volume of transactional data daily; however, raw data alone does not provide meaningful insight into performance. Management requires visibility into sales trends, product demand, customer purchasing behavior, payment preferences, and marketing channel effectiveness to make informed business decisions.

This analysis was conducted to answer key business questions such as:

- How many orders have been processed?
- Which products generate the highest sales and revenue?
- Which payment methods are most frequently used by customers?
- Which referral channels contribute the most revenue?
- What is the average value of customer orders?
- Are there any unusually high-value transactions that require further investigation?
- Which operational areas present opportunities for revenue growth and process improvement?

By answering these questions, the business can better understand its sales performance, customer behavior, and revenue drivers.

3. Objectives

The primary objectives of this analysis were to:

- Analyze overall order volume and transaction activity.
- Evaluate revenue performance across products, payment methods, referral sources, and order statuses.
- Identify top-performing products based on order frequency and revenue generation.
- Assess customer payment preferences and their contribution to revenue.
- Evaluate the effectiveness of referral channels in driving sales.

- Determine average order values and purchasing patterns.
- Identify unusual or high-value transactions through outlier analysis.
- Highlight the highest and lowest transaction values within the dataset.

4. SQL Techniques Used

The following SQL techniques were utilized throughout the analysis:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

These techniques enabled efficient extraction, filtering, aggregation, and summarization of transactional data to support business decision-making.

5. Key Findings

Order Volume Analysis

- A total of **1200** orders were analyzed.
- Order status analysis showed that **Cancelled Status** was the most frequent order outcome.
- Product order counts identified the most frequently purchased products within the business.

Product Performance Analysis

- **Laptop** generated the highest revenue among all products.
- **Printer** recorded the highest number of orders.
- Product-level average order value analysis highlighted products generating the highest customer spend per transaction.

Revenue Analysis

- Total revenue generated from delivered orders amounted to **\$242,600.32**.
- Revenue distribution varied across products, payment methods, referral channels, and months.
- Monthly revenue analysis identified periods of stronger sales performance and customer demand.

Payment Method Analysis

- **Digital payment method** (Online) was the most frequently used payment option.
- **Digital payment method** (Online) generated the highest revenue contribution.
- Customer spending patterns differed across payment methods based on average order value.

Referral Source Analysis

- **Email** generated the highest revenue contribution in terms of fulfilled orders, while **Instagram** generated the highest order.
- Referral channel performance revealed the most effective customer acquisition sources.

Customer Purchasing Behavior

- Average purchase quantities varied across products and payment methods.
- Certain products demonstrated stronger purchasing demand based on average quantity ordered.

Outlier and High-Value Order Analysis

- Outlier analysis identified transactions with unusually high order values relative to the overall dataset.
- The Top 10 highest-value orders were isolated for further investigation.
- These transactions may represent premium purchases, bulk orders, or exceptional customer spending behavior.

Order Value Analysis

- The highest total order value recorded was **\$3456.39**.
- The lowest total order value recorded was **\$11.39**.
- The highest unit price recorded was **\$699.92**.
- The lowest unit price recorded was **\$11.39**.

6. Business Impact

This analysis provides management with actionable insights into sales performance and customer behavior. The findings can be used to:

- Focus marketing efforts on the most effective referral channels.
- Promote products that generate the highest revenue and customer demand.
- Improve payment experiences based on customer preferences.
- Monitor and investigate unusual transactions.
- Support strategic decisions aimed at increasing revenue and operational efficiency.
- Enhance customer acquisition and retention strategies.

- Improve customer satisfaction strategies to reduce pending, returned, and cancelled orders.

7. Conclusion

The analysis provided valuable insight into order activity, revenue performance, customer purchasing behavior, and sales channel effectiveness. Product-level and payment-level evaluations identified key revenue drivers, while referral source analysis highlighted the most effective acquisition channels. Outlier detection and high-value order analysis further enhanced understanding of exceptional transactions within the dataset.

Overall, SQL aggregation and filtering techniques enabled efficient exploration of transactional data and transformed raw order data into meaningful business intelligence. The findings provide a foundation for improving sales performance, optimizing marketing efforts, and supporting data-driven business decisions.